

Neha Patel

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Education

Charotar University of Science & Technology (CHARUSAT)

2023 – 2027

Bachelor of Technology in Information Technology – CGPA: 7.64/10.0

Anand, Gujarat

Technical Skills

Programming Languages: Python, JavaScript, C, SQL

AI/ML Technologies: Machine Learning, Neural Networks, Natural Language Processing, Supervised Learning, Model Training

Web Development: React, Full-Stack Development, REST APIs

Frameworks & Tools: Node.js, Git, Docker, Data Preprocessing, Model Evaluation

Database & Cloud: SQL, Cloud Deployment, AWS (S3, Lambda, API Gateway, IAM, CloudWatch)

Other Skills: Cybersecurity, Ethical Hacking, Data Analysis, Graphic Design, Algorithm Design

Experience

Tech Elecon Pvt. Ltd

May 2025 – June 2025

Web Development Intern

Hybrid, India

- Completed a four week internship focus on a front end web development using modern frameworks.
- Develop responsive and user friendly web interfaces.
- Worked with the Javascript frameworks and component based architecture.
- Improved skills in frontend design, debugging and performance optimization.

Projects

Traffic Light Time Prediction | Python, OpenCV, Machine Learning

December 2025

- Developed an intelligent traffic system to predict signal phase duration (red/green) using historical traffic detained real time vehicle density.
- Perform data collection and preprocessing using Pandas and NumPy including feature engineering time-of-day and traffic flow.
- Trained regression models to estimate signal timing in second and evaluated performance using error metrics
- Integrated a computer vision model (Open CV/YOLO) to count vehicles from traffic video stream as real-time time traffic.

Cloud-Based Document Processing & Management System | AWS, React, Python

January 2026

- Developed a serverless document processing platform using Amazon S3, Lambda, API Gateway, DynamoDB, and SQS for scalable event-driven processing.
- Implemented automated document upload and processing workflows using S3 event triggers and AWS Lambda.
- Designed REST APIs with API Gateway and integrated DynamoDB to track document metadata and processing status.
- Configured IAM least-privilege access and CloudWatch monitoring/logging to improve application security and observability.
- Built a responsive React interface for document upload, status tracking, and processed-result retrieval.

ALP – Autonomous Learning Partner (Private AI Assistant) | Python, LangChain, HuggingFace, FastAPI, AWS EC2

April 2026

- Designed and developed a privacy-first AI assistant using a Retrieval-Augmented Generation (RAG) architecture.
- Implemented a local LLM pipeline ensuring zero data leakage and offline-first processing.
- Built document processing pipeline using PyMuPDF with text chunking and embedding generation.
- Integrated ChromaDB vector database for high-speed semantic search and retrieval.
- Developed REST APIs using FastAPI for efficient backend communication.

- Achieved accurate, context-aware responses from large PDF datasets.

Achievements & Activities

- **Certifications:** The Ultimate Job Ready Data Science Course, Ethical Hacking Essentials (EC-Council), Foundations of Cybersecurity (Google), AWS Academy Graduate

Research publication

Transponder- Free emergency vehicle preemption and adaptive signal control for heterogeneous Indian urban Traffic using DQN in YOLO V8

- Springer Conference – ICT4SD 2026 (11th International ICT Summit & Awards) Goa, India
- Research paper accepted for publication in Springer conference proceedings at ICT4SD 2026.
- Proposed a camera-based adaptive traffic signal control system using YOLOv8 for real-time vehicle detection and Deep Q-Network (DQN) for dynamic signal optimization.
- Eliminated need for transponders and inductive loop sensors, enabling deployment using existing CCTV infrastructure.
- Achieved 13.6%–16.8% reduction in average vehicle delay and 91.3% faster emergency vehicle clearance compared to traditional fixed-time systems.
- Implemented and evaluated system using SUMO simulation (v1.24.0) on real-world traffic data from Ahmedabad.
- Maintained >89% detection accuracy under adverse conditions (rain, fog, night).
- Ensured real-time performance (≈ 41 ms response time) suitable for edge deployment.

